



Computer System and Architecture

Machine Problem

MIDTERM

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List the steps that you perform to come up with the program.

STEP 1: I read the instructions first and then examined the slam book design sample then used it as my reference for my own.

STEP 2: I manually designed the layout and ascii art in Notepad before writing any assembly code. (It took me about 7 hours to finish it)

STEP 3: I modified some of the categories so I could make it more personalized

STEP 4: After I finalized the entire structure I added each section with my infos.

STEP 5: I adjusted the spacing, borders, the ascii art, and text alignment to make it perfect and no errors.

STEP 6: After completing the design, I turned each row of the slambook into Assembly string data then of course added the length value for each.

STEP 7: I used a reusable print screen procedure so it will display each string using `sys_write`.

STEP 8: I added the instructions in `_start` for each row so it will print in order.

STEP 9: I added a border loop, so it won't have to be manually.

STEP 10: I assembled and tested the program

STEP 11: I fixed some minor issues during assembly and testing and then finalized the program and confirmed that the final version is all good.

Paste the source code and explain the functionalities and take a screenshot of the final output.

FUNCTIONALITY OF THE PROGRAM:

- The program is an ascii art personal slambook which I created with NASM 32-bit Assembly. It displays my personal information, interests, hobbies, and goals in a structured console based interface.
- The **.data** section contains all of the texts and ascii artwork that's displayed by the program. Each row of the slambook is stored as a separate string using the `db` directive.



- ### Source Code:

```

; =====
; MIDTERM PROJECT & EXAM - ASCII SLAMBOOK
; NASM 32-bit Linux
; by: Angelika Joy Abendanio
; =====

section .data
    title db "MY PROFILE - ASCII SLAMBOOK", 10
    titleLen equ $ - title

    row1 db
"=====+
=====+", 10
    row1Len equ $ - row1
    row2 db "|
|", 10
    row2Len equ $ - row2
    row3 db "|
|", 10
    row3Len equ $ - row3
    row4 db "|
| | | | | | | | | | |", 10
    row4Len equ $ - row4
    row5 db "|
| | | | | | | | | | |", 10
    row5Len equ $ - row5
    row6 db "|
| | | | | | | | | | |", 10
    row6Len equ $ - row6
    row7 db "|
|_| | |_| | | | |_| |_| \_| \_| \_| /_| |", 10
    row7Len equ $ - row7
    row8 db "|
|", 10
    row8Len equ $ - row8
    row9 db "|
|", 10
    row9Len equ $ - row9
    row10 db
"=====+
=====+", 10

```



```
row10Len equ $ - row10
row11 db "
", 10
row11Len equ $ - row11
row12 db "+----- ABOUT ME -----
-----+", 10
row12Len equ $ - row12
row13 db "|
", 10
row13Len equ $ - row13
row14 db "|    NAME      : ANGELIKA JOY ABENDANIO
", 10
row14Len equ $ - row14
row15 db "|    TYPE      : STUDENT / CREATIVE / CURIOUS
", 10
row15Len equ $ - row15
row16 db "|    STATUS    : LOCKED IN FOR MIDTERMS XD
", 10
row16Len equ $ - row16
row17 db "|
", 10
row17Len equ $ - row17
row18 db "|    I like exploring technology, games, stories,
art, and the things    |", 10
row18Len equ $ - row18
row19 db "|    that make me curious.
", 10
row19Len equ $ - row19
row20 db "|
", 10
row20Len equ $ - row20
row21 db "+-----+-----
-----+", 10
row21Len equ $ - row21
row22 db "|
", 10
row22Len equ $ - row22
row23 db "|          M Y P E T C A T          |          M Y H
I G H L I G H T S          |", 10
row23Len equ $ - row23
row24 db "|
", 10
row24Len equ $ - row24
row25 db "|          + BIGGEST
ACHIEVEMENT          |", 10
row25Len equ $ - row25
```



```
row26 db "|                /\_/\                |    Pursuing the
things I love, actually      |", 10
row26Len equ $ - row26
row27 db "|                (=^.^=)                |    thriving in
them, and enjoying them.    |", 10
row27Len equ $ - row27
row28 db "|                >    <    \\\                |
|", 10
row28Len equ $ - row28
row29 db "|                / | | \_//                |
|", 10
row29Len equ $ - row29
row30 db "|                | ( _ ) ( _ ) | /                |    + HAPPIEST
MOMENT                      |", 10
row30Len equ $ - row30
row31 db "|                |                |    Visiting a
Gundam Base in South Korea  |", 10
row31Len equ $ - row31
row32 db "|                |                |    for the first
time.                      |", 10
row32Len equ $ - row32
row33 db "|                T O F U                |
|", 10
row33Len equ $ - row33
row34 db "|                M E R L                |
|", 10
row34Len equ $ - row34
row35 db "|                |                |
|", 10
row35Len equ $ - row35
row36 db "+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+
row36Len equ $ - row36
row37 db "
", 10
row37Len equ $ - row37
row38 db "
", 10
row38Len equ $ - row38
row39 db "+-----+-----+-----+-----+-----+-----+ MY FAVORITES -----
-----+-----+-----+-----+-----+-----+-----+
row39Len equ $ - row39
row40 db "|
|", 10
row40Len equ $ - row40
row41 db "|    COLOR                : BLUE + GREEN
|", 10
```



6



7



```
row72Len equ $ - row72
row73 db "
", 10
row73Len equ $ - row73
row74 db "+----- MY HOBBIES -----
-----+", 10
row74Len equ $ - row74
row75 db "|
", 10
row75Len equ $ - row75
row76 db "| [1] GUNPLA
", 10
row76Len equ $ - row76
row77 db "| I collect UC Timeline Gundam action figures
", 10
row77Len equ $ - row77
row78 db "|
", 10
row78Len equ $ - row78
row79 db "| [2] VIDEO GAMES
", 10
row79Len equ $ - row79
row80 db "| I play open world games especially games
that let me explore. |", 10
row80Len equ $ - row80
row81 db "|
", 10
row81Len equ $ - row81
row82 db "| [3] ACTION FIGURES
", 10
row82Len equ $ - row82
row83 db "| I also collect military figures
particularly GI JOES. |", 10
row83Len equ $ - row83
row84 db "|
", 10
row84Len equ $ - row84
row85 db "| [4] BINGE WATCHING
", 10
row85Len equ $ - row85
row86 db "| Movies, anime, K-dramas, and anything
interesting. |", 10
row86Len equ $ - row86
row87 db "|
", 10
row87Len equ $ - row87
```



```
row88 db "|    [5] DRAWING + WRITING
|", 10
row88Len equ $ - row88
row89 db "|          Sometimes I draw. Sometimes I write.
Usually when I'm bored.          |", 10
row89Len equ $ - row89
row90 db "|
|", 10
row90Len equ $ - row90
row91 db "+-----"
-----+
row91Len equ $ - row91
row92 db "
", 10
row92Len equ $ - row92
row93 db "
", 10
row93Len equ $ - row93
row94 db "+----- MY CAREER INTEREST ---
-----+
row94Len equ $ - row94
row95 db "|
|", 10
row95Len equ $ - row95
row96 db "|    I am very interested in AI, especially its
developing philosophy.          |", 10
row96Len equ $ - row96
row97 db "|
|", 10
row97Len equ $ - row97
row98 db "|    I am also interested in technology such as:
|", 10
row98Len equ $ - row98
row99 db "|
|    |          |", 10
row99Len equ $ - row99
row100 db "|          > ARTIFICIAL INTELLIGENCE
| [Hello] |          |", 10
row100Len equ $ - row100
row101 db "|          > VIDEO GAMES
|          |          |", 10
row101Len equ $ - row101
row102 db "|          > CREATIVITY
/-----/          |", 10
row102Len equ $ - row102
row103 db "|          > TECHNOLOGY
/          /          |", 10
```



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```
row119 db "|                |          [ BOOK ]          |
|", 10
row119Len equ $ - row119
row120 db "|                |          _____          |
|", 10
row120Len equ $ - row120
row121 db "|                |          /_____/|          |
|", 10
row121Len equ $ - row121
row122 db "|                |          | BOOK |          |
|", 10
row122Len equ $ - row122
row123 db "|                |          |_____|/          |
|", 10
row123Len equ $ - row123
row124 db "|                +-----+
|", 10
row124Len equ $ - row124
row125 db "|
|", 10
row125Len equ $ - row125
row126 db "+-----+
-----+ ", 10
row126Len equ $ - row126
row127 db "
", 10
row127Len equ $ - row127
row128 db "
", 10
row128Len equ $ - row128
row129 db "+----- QUOTE -----
-----+ ", 10
row129Len equ $ - row129
row130 db "|
|", 10
row130Len equ $ - row130
row131 db '|                "THE WIND IS RISING! WE MUST TRY
TO LIVE."                |', 10
row131Len equ $ - row131
row132 db "|                -
Paul Valery                |", 10
row132Len equ $ - row132
row133 db "|
|", 10
row133Len equ $ - row133
row134 db "+-----+
-----+ ", 10
```



```
    row134Len equ $ - row134
    row135 db "
", 10
    row135Len equ $ - row135
    row136 db "
", 10
    row136Len equ $ - row136
    row137 db
"+=====
=====+"", 10
    row137Len equ $ - row137
    row138 db "|
", 10
    row138Len equ $ - row138
    row139 db
"+=====
=====+"", 10
    row139Len equ $ - row139

    borderChar db "="
    newline db 10

section .text
    global _start

_start:
    ; Prints the complete slambook row by row.
    mov ecx, row1
    mov edx, row1Len
    call printString
    mov ecx, row2
    mov edx, row2Len
    call printString
    mov ecx, row3
    mov edx, row3Len
    call printString
    mov ecx, row4
    mov edx, row4Len
    call printString
    mov ecx, row5
    mov edx, row5Len
    call printString
    mov ecx, row6
    mov edx, row6Len
    call printString
    mov ecx, row7
    mov edx, row7Len
```



```
call printString
mov ecx, row8
mov edx, row8Len
call printString
mov ecx, row9
mov edx, row9Len
call printString
mov ecx, row10
mov edx, row10Len
call printString
mov ecx, row11
mov edx, row11Len
call printString
mov ecx, row12
mov edx, row12Len
call printString
mov ecx, row13
mov edx, row13Len
call printString
mov ecx, row14
mov edx, row14Len
call printString
mov ecx, row15
mov edx, row15Len
call printString
mov ecx, row16
mov edx, row16Len
call printString
mov ecx, row17
mov edx, row17Len
call printString
mov ecx, row18
mov edx, row18Len
call printString
mov ecx, row19
mov edx, row19Len
call printString
mov ecx, row20
mov edx, row20Len
call printString
mov ecx, row21
mov edx, row21Len
call printString
mov ecx, row22
mov edx, row22Len
call printString
mov ecx, row23
```



```
mov edx, row23Len
call printString
mov ecx, row24
mov edx, row24Len
call printString
mov ecx, row25
mov edx, row25Len
call printString
mov ecx, row26
mov edx, row26Len
call printString
mov ecx, row27
mov edx, row27Len
call printString
mov ecx, row28
mov edx, row28Len
call printString
mov ecx, row29
mov edx, row29Len
call printString
mov ecx, row30
mov edx, row30Len
call printString
mov ecx, row31
mov edx, row31Len
call printString
mov ecx, row32
mov edx, row32Len
call printString
mov ecx, row33
mov edx, row33Len
call printString
mov ecx, row34
mov edx, row34Len
call printString
mov ecx, row35
mov edx, row35Len
call printString
mov ecx, row36
mov edx, row36Len
call printString
mov ecx, row37
mov edx, row37Len
call printString
mov ecx, row38
mov edx, row38Len
call printString
```



```
mov ecx, row39
mov edx, row39Len
call printString
mov ecx, row40
mov edx, row40Len
call printString
mov ecx, row41
mov edx, row41Len
call printString
mov ecx, row42
mov edx, row42Len
call printString
mov ecx, row43
mov edx, row43Len
call printString
mov ecx, row44
mov edx, row44Len
call printString
mov ecx, row45
mov edx, row45Len
call printString
mov ecx, row46
mov edx, row46Len
call printString
mov ecx, row47
mov edx, row47Len
call printString
mov ecx, row48
mov edx, row48Len
call printString
mov ecx, row49
mov edx, row49Len
call printString
mov ecx, row50
mov edx, row50Len
call printString
mov ecx, row51
mov edx, row51Len
call printString
mov ecx, row52
mov edx, row52Len
call printString
mov ecx, row53
mov edx, row53Len
call printString
mov ecx, row54
mov edx, row54Len
```



```
call printString
mov ecx, row55
mov edx, row55Len
call printString
mov ecx, row56
mov edx, row56Len
call printString
mov ecx, row57
mov edx, row57Len
call printString
mov ecx, row58
mov edx, row58Len
call printString
mov ecx, row59
mov edx, row59Len
call printString
mov ecx, row60
mov edx, row60Len
call printString
mov ecx, row61
mov edx, row61Len
call printString
mov ecx, row62
mov edx, row62Len
call printString
mov ecx, row63
mov edx, row63Len
call printString
mov ecx, row64
mov edx, row64Len
call printString
mov ecx, row65
mov edx, row65Len
call printString
mov ecx, row66
mov edx, row66Len
call printString
mov ecx, row67
mov edx, row67Len
call printString
mov ecx, row68
mov edx, row68Len
call printString
mov ecx, row69
mov edx, row69Len
call printString
mov ecx, row70
```



```
mov edx, row70Len
call printString
mov ecx, row71
mov edx, row71Len
call printString
mov ecx, row72
mov edx, row72Len
call printString
mov ecx, row73
mov edx, row73Len
call printString
mov ecx, row74
mov edx, row74Len
call printString
mov ecx, row75
mov edx, row75Len
call printString
mov ecx, row76
mov edx, row76Len
call printString
mov ecx, row77
mov edx, row77Len
call printString
mov ecx, row78
mov edx, row78Len
call printString
mov ecx, row79
mov edx, row79Len
call printString
mov ecx, row80
mov edx, row80Len
call printString
mov ecx, row81
mov edx, row81Len
call printString
mov ecx, row82
mov edx, row82Len
call printString
mov ecx, row83
mov edx, row83Len
call printString
mov ecx, row84
mov edx, row84Len
call printString
mov ecx, row85
mov edx, row85Len
call printString
```



```
mov ecx, row86
mov edx, row86Len
call printString
mov ecx, row87
mov edx, row87Len
call printString
mov ecx, row88
mov edx, row88Len
call printString
mov ecx, row89
mov edx, row89Len
call printString
mov ecx, row90
mov edx, row90Len
call printString
mov ecx, row91
mov edx, row91Len
call printString
mov ecx, row92
mov edx, row92Len
call printString
mov ecx, row93
mov edx, row93Len
call printString
mov ecx, row94
mov edx, row94Len
call printString
mov ecx, row95
mov edx, row95Len
call printString
mov ecx, row96
mov edx, row96Len
call printString
mov ecx, row97
mov edx, row97Len
call printString
mov ecx, row98
mov edx, row98Len
call printString
mov ecx, row99
mov edx, row99Len
call printString
mov ecx, row100
mov edx, row100Len
call printString
mov ecx, row101
mov edx, row101Len
```



```
call printString
mov ecx, row102
mov edx, row102Len
call printString
mov ecx, row103
mov edx, row103Len
call printString
mov ecx, row104
mov edx, row104Len
call printString
mov ecx, row105
mov edx, row105Len
call printString
mov ecx, row106
mov edx, row106Len
call printString
mov ecx, row107
mov edx, row107Len
call printString
mov ecx, row108
mov edx, row108Len
call printString
mov ecx, row109
mov edx, row109Len
call printString
mov ecx, row110
mov edx, row110Len
call printString
mov ecx, row111
mov edx, row111Len
call printString
mov ecx, row112
mov edx, row112Len
call printString
mov ecx, row113
mov edx, row113Len
call printString
mov ecx, row114
mov edx, row114Len
call printString
mov ecx, row115
mov edx, row115Len
call printString
mov ecx, row116
mov edx, row116Len
call printString
mov ecx, row117
```



```
mov edx, row117Len
call printString
mov ecx, row118
mov edx, row118Len
call printString
mov ecx, row119
mov edx, row119Len
call printString
mov ecx, row120
mov edx, row120Len
call printString
mov ecx, row121
mov edx, row121Len
call printString
mov ecx, row122
mov edx, row122Len
call printString
mov ecx, row123
mov edx, row123Len
call printString
mov ecx, row124
mov edx, row124Len
call printString
mov ecx, row125
mov edx, row125Len
call printString
mov ecx, row126
mov edx, row126Len
call printString
mov ecx, row127
mov edx, row127Len
call printString
mov ecx, row128
mov edx, row128Len
call printString
mov ecx, row129
mov edx, row129Len
call printString
mov ecx, row130
mov edx, row130Len
call printString
mov ecx, row131
mov edx, row131Len
call printString
mov ecx, row132
mov edx, row132Len
call printString
```



```
mov ecx, row133
mov edx, row133Len
call printString
mov ecx, row134
mov edx, row134Len
call printString
mov ecx, row135
mov edx, row135Len
call printString
mov ecx, row136
mov edx, row136Len
call printString
mov ecx, row137
mov edx, row137Len
call printString
mov ecx, row138
mov edx, row138Len
call printString
mov ecx, row139
mov edx, row139Len
call printString

; Exit program
mov eax, 1
mov ebx, 0
int 0x80

; Prints one string.
printString:
    mov eax, 4
    mov ebx, 1
    int 0x80
    ret

; Prints the reusable 78 column border.
printBorder:
    mov esi, 78

borderLoop:
    push esi
    mov eax, 4
    mov ebx, 1
    mov ecx, borderChar
    mov edx, 1
    int 0x80
    pop esi
    dec esi
```



```
jnz borderLoop

mov eax, 4
mov ebx, 1
mov ecx, newline
mov edx, 1
int 0x80
ret
```

Output:

```
PS C:\Users\AJ Abendanio\Documents\NASM\hjtnasm> docker compose run --rm nasm
Container hjtnasm-nasm-run-4271baa0b314 Creating
Container hjtnasm-nasm-run-4271baa0b314 Created
root@6a67c12ec061:/work# nasm -f elf32 midterm.asm -o midterm.o
root@6a67c12ec061:/work# ld -m elf_i386 midterm.o -o midterm
root@6a67c12ec061:/work# ./midterm

=====
              MY MIDTERM
=====

----- ABOUT ME -----
NAME      : ANGELIKA JOY ABENDANIO
TYPE      : STUDENT / CREATIVE / CURIOUS
STATUS    : LOCKED IN FOR MIDTERMS XD

I like exploring technology, games, stories, art, and the things
that make me curious.

=====
MY PET CAT      MY HIGHLIGHTS
=====

      /\
      (=^.^=)
      > <  \
      / | \  /
      |()()|/

      T O F U
      M E R L

+ BIGGEST ACHIEVEMENT
Pursuing the things I love, actually
thriving in them, and enjoying them.

+ HAPPIEST MOMENT
Visiting a Gundam Base in South Korea
for the first time.
```



MY FAVORITES	
COLOR	: BLUE + GREEN
MUSIC	: JAZZ + CLASSICAL
FOOD	: FISH AND CHIPS !!!
MOVIES / FILMS	
Studio Ghibli is a favorite, especially THE WIND RISES. <3	
GAMES	GUNDAM / GUNPLA
RED DEAD REDEMPTION 2	
~700 HOURS	
STARDEW VALLEY	
GTA V	
THE WITCHER 3	
CYBERPUNK 2077	
OTHER OPENWORLD GAMES...	I love the UC Timeline Gundams and building Gunpla.



----- MY HOBBIES -----

[1] GUNPLA
I collect UC Timeline Gundam action figures

[2] VIDEO GAMES
I play open world games especially games that let me explore.

[3] ACTION FIGURES
I also collect military figures particularly GI JOES.

[4] BINGE WATCHING
Movies, anime, K-dramas, and anything interesting.

[5] DRAWING + WRITING
Sometimes I draw. Sometimes I write. Usually when I'm bored.

----- MY CAREER INTEREST -----

I am very interested in AI, especially its developing philosophy.

I am also interested in technology such as:

> ARTIFICIAL INTELLIGENCE

> VIDEO GAMES

> CREATIVITY

> TECHNOLOGY

[Hello]

/-----/

/-----/

I enjoy thinking about where technology is going and what it means.

MIDTERM EXAM

24



```
+----- WHAT I WANT TO DO NEXT -----+
|
|   READ MORE
|
|   I want to read lots of books and explore different worlds,
|   ideas, and stories. I also want to spend more time browsing
|   libraries and discovering books that catch my attention.
|
|   +-----+
|   | [ BOOK ] |
|   |         |
|   |  /      |
|   | BOOK   |
|   |  \      |
|   |         |
|   +-----+
|
+-----+

+----- QUOTE -----+
|
|   "THE WIND IS RISING! WE MUST TRY TO LIVE."
|   - Paul Valery
|
+-----+

+=====+
|
+=====+
root@6a67c12ec061:/work#
```